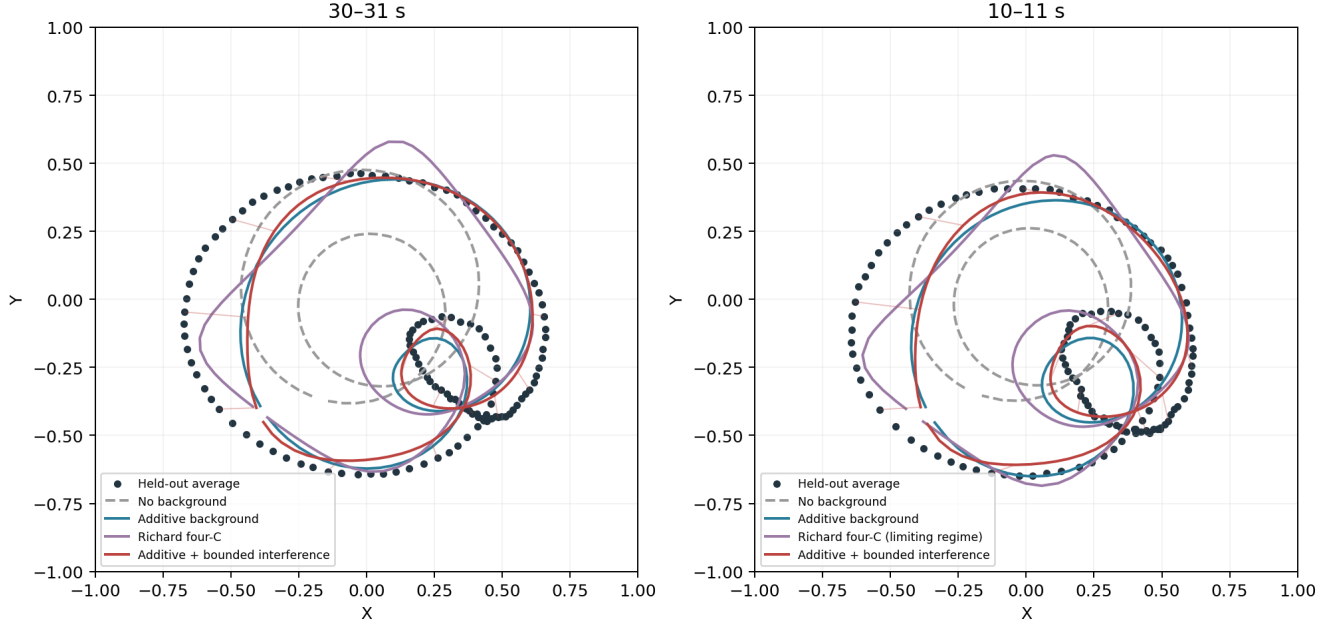


Corrected final-reference correspondence

6 October 2026

Correction to earlier reports. The final channel-averaged trajectory was not uniformly spaced in anisotropy, although its constituent cycles had equal-guide-arc bins. Matching that final reference at uniform index fractions was incorrect. This report supersedes the previous equal-arc forward-fit comparisons, including the forward part of the complex/inverse-circle report. The inverse-circle collapse counterexample is independent of this error.

Corrected correspondence: actual final-reference arc positions



Observed points are intentionally unchanged and therefore remain nonuniformly spaced. The model is now evaluated at their actual training-reference arc fractions. Thin red segments join selected corresponding observed and predicted points; no nearest-neighbour matching is used.

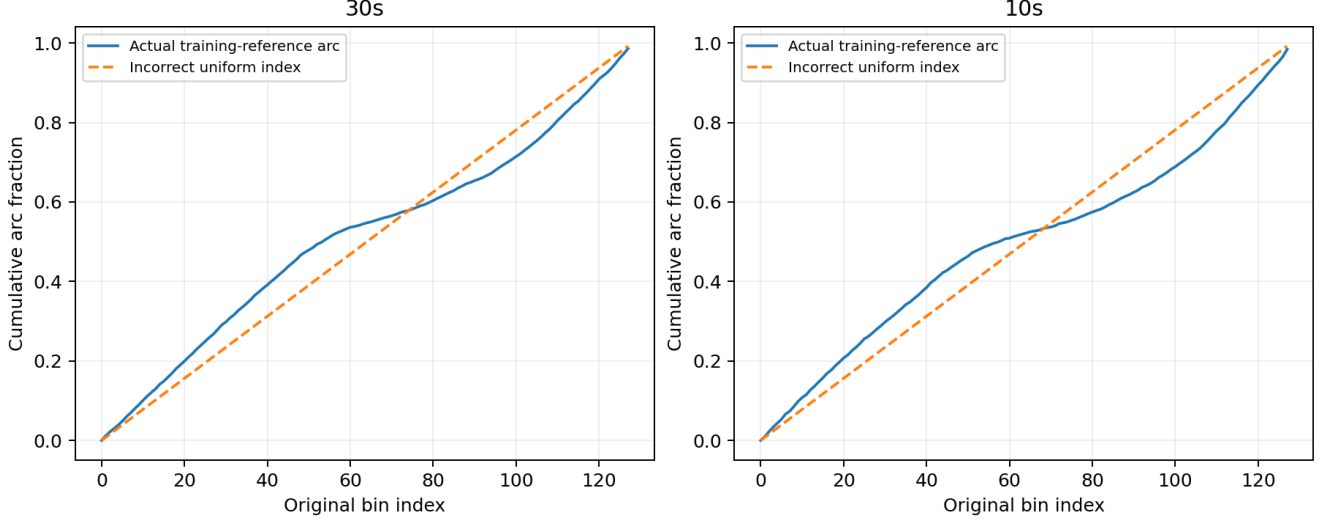
Window	Model	Channel RMS	XY RMS	BG/ $T_{\max}$	θ range
30s	None	0.08281	0.41166	0.0%	29.0–44.4°
30s	Additive	0.02902	0.11611	37.3%	29.4–90.0°
30s	Richard four-C	0.03930	0.15151	—	28.9–89.0°
30s	Additive + interference	0.02670	0.10775	45.7%	30.3–90.0°
10s	None	0.08242	0.41597	0.0%	30.3–41.9°
10s	Additive	0.03333	0.13886	40.3%	33.8–89.8°
10s	Richard four-C*	0.03973	0.15963	—	33.4–89.0°
10s	Additive + interference	0.02987	0.12507	43.8%	34.7–90.0°

*Richard's 10 s row is a limiting regime with vanishing signal brightness and growing coefficients; it is not a stable finite-coefficient correction. No physical background power is inferred from Richard's approximation. All angular ranges are conditional model outputs, not independently validated angles.

What changed, and how it was checked

For each final training track, interpolate the four intensities together along the closed, ordered list of bins. Calculate X/Y from this channel interpolant, integrate its arc length, and record the fraction u_k at each original bin. A candidate full measured-model curve is then evaluated at fractions $(\delta \pm u_k) \bmod 1$, with one fitted cyclic offset and both directions tested.

The held-out track uses exactly the training-derived u_k , offset, direction and fitted physical parameters. It is not independently reparameterized. This preserves every original observed channel value and the original equal-bin intensity weighting. It is not a new uniformly reweighted reference; uniform resampling would additionally change the objective weights.



The intensity objective remains

$$J = \sum_{k,i} \left[\frac{I_{i,k}^{\text{pred}} - I_{i,k}^{\text{obs}}}{T_k^{\text{obs}}} \right]^2.$$

Three cone parameters, one brightness scale and one starting offset are fitted, plus correction parameters. No 16-parameter phase deformation. Optics, fixed gains/T, NA 0.38–1.3 and cycle selection remain unchanged (304/217 traversals). Training and test averages use alternate traversals.

Numerical check. A deliberately nonuniformly sampled synthetic model curve produces channel RMS 0.11364 when incorrectly matched by uniform indices. The corrected coordinate method reduces this to 1.54×10^{-5} , with the remaining difference due to finite reference interpolation. Increasing reference interpolation subdivisions from 32 to 128 changes fractions by less than 10^{-11} . The selected finite non-Richard fits are refined at 8192 simulated segments; comparison at 16384 changes intensities by less than $4 \times 10^{-8} T_{\text{max}}$.

Search and comparisons. Each ordinary model/window used 18–19 coarse starts; the complex model used 24, across both directions, including nested zero/additive seeds and varied geometry. Selected none/additive/complex fits converge and beat their nested training baselines. Local search is not a global optimality certificate. Earlier flexible-phase controls use the same unchanged observations and still have lower held-out channel errors for the three original model families. That does not establish better physical angles. The coordinate repair changes recovered geometry materially even where the score changes modestly.

Brightness constraint and the remaining Richard degeneracy

For the fixed circular-excitation model,

$$S_i = a_0^2 \sin^2 \theta q_i(\theta, \phi), \quad \sum_i S_i = 4a_0^2 \sin^2 \theta (A_F + B_F \sin^2 \theta).$$

A sphere-circle inversion can use this total-intensity law at the recovered/projected orientations, plus the channel pair-sum condition. It must not choose an independent brightness per point. Time order supports phi unwrapping but does not by itself determine theta reflection branches at an equator crossing; those choices require continuity and physical-consistency checks. This report repairs correspondence only; it does not add new inverse-fit constraints.

Richard's model has another potential degeneracy even in a forward cone fit. Write $S_i = a_0^2 f_i$, then

$$I_i = a_0^2 f_i + (a_0 C_i) \sqrt{f_i} = \alpha f_i + K_i \sqrt{f_i}, \quad \alpha = a_0^2, \quad K_i = a_0 C_i.$$

For positive alpha this is exactly the same model, with no additional degrees of freedom. The optimizer can reduce alpha toward zero while keeping K finite, which sends C to infinity. The original 10 s coefficient fit exhausted its refinement budget along this direction. Reparameterizing and searching nine starts converges to $\alpha \approx 6.7 \times 10^{-15}$, confirming the limiting tendency. Its predicted curve is well defined as a limit, but its separated signal/interference amplitudes are not a stable correction. The earlier numerical lower bound on brightness merely truncates this direction.

For 30 s the selected reparameterized solution has finite $\alpha \approx 0.01063$. Its physical adequacy still requires independent constraints. The bounded additive-plus-interference model instead uses

$$I_i = S_i + B_i + 2\rho_i \sqrt{B_i S_i}, \quad |\rho_i| \leq 1, \quad B_{\text{total}} \leq 0.95 \min T_{\text{train}}.$$

A finite background budget bounds its effective coefficients. These channel-wise bounds are necessary physical restrictions but do not establish realization by one common optical field.

Interpretation. The correspondence bug is repaired. The bounded-interference model now gives the smallest errors among these four corrected fixed-correspondence fits, but substantial mismatch remains. The result supports testing intensity/physical-budget constraints in the inverse-circle approach next, without claiming that current fitted angles are established or that flexible matching is physically superior.

Scripts, input-coordinate arrays, fit trials, predictions, resolution checks and Richard limiting analysis are saved in `model_comparison/corrected_arc/`. No commit or push.